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Carleman approximation based adaptive optimal control design of nonlinear systems: A three-phase policy iteration approach

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ABSTRACT

As one efficient algorithm in adaptive dynamic programming to address adaptive optimal control problems, policy iteration always involves an initial admissible control guess during iteration. Such an initialization process is nontrivial, especially for unstable systems or when system dynamics are totally unknown. To circumvent this limitation, this paper develops a novel adaptive optimal controller design scheme for continuous-time nonlinear systems through neural network-based policy iteration. By employing Carleman linearization, we lift the nonlinear model into a bilinear form, which allows the optimal feedback control to be derived from a state-dependent Riccati (SDR) equation rather than the Hamilton-Jacobi-Bellman equation. To solve the SDR equation under unknown system parameters, we leverage a homotopy-based three-phase policy iteration algorithm that learns the optimal control protocol directly from state and input measurements. In comparison to existing results, the proposed algorithm has the ability to automatically update the initial parameters and specifies the same basis functions of neural networks to approximate all unknown functions, which gives rise to computational improvement. Two case studies are provided to validate our methodology.

1. Introduction

Optimal control refers to designing a control protocol that minimizes (or maximizes) a specified performance criterion (Peng et al., 2026; Qin et al., 2026b,d; Shen et al., 2024; Zhang et al., 2025). This mathematical formulation is capable of leading to best possible behavior for closed-loop plants and is thus leveraged to address numerous real-world engineering problems, for instance, only to name a few, safe control of modular robot manipulators (Dong et al., 2025), adaptive cruise control of electric vehicles (Hattori et al., 2022), optimal operational control of mineral processes (Noone et al., 2023), and formulation control of tool affordance (Ti et al., 2024). According to dynamic programming and Pontryagin's principle (Bertsekas, 2005; Sassano & Astolfi, 2020), the optimal control solution with respect to nonlinear systems could be obtained by solving the so-called Hamilton-Jacobi-Bellman (HJB) equation, which is simplified as an algebraic Riccati equation for linear systems. It should be noted that addressing nonlinear optimal control problems presents at least two significant difficulties. Firstly, the corresponding HJB equation is a nonlinear partial differential equation and generally does not possess an analytical solution. Secondly, the optimal controller designed based on model parameters cannot be performed

when implemented in real-world scenarios with unavailable system dynamics.

Adaptive dynamic programming (ADP), a new branch of reinforcement learning, is a computational intelligence technique which iteratively learns an optimal control law under unknown environment (Ganie & Jagannathan, 2025; Qin et al., 2026a; Shen et al., 2026; Wang et al., 2024; Yue et al., 2025). Differing from classical reinforcement learning, ADP can be leveraged to tackle decision-making problems in dynamical systems represented by differential equations, where both states and actions exist in continuous spaces (Sutton & Barto, 2018). By means of data measurements, ADP has become a valuable candidate to deal with continuous-time adaptive optimal control problems in the absence of the exact knowledge of system parameters (Cui et al., 2024; Jiang et al., 2025; Zhao et al., 2023, 2025). Using Galerkin's approximation, Vrabie and Lewis (2009) and Luo et al. (2014) respectively developed on-policy and off-policy ADP algorithms for solving HJB equation with unknown drift dynamic and completely unknown dynamics. Possieri and Sassano (2023) further presented a novel critic-identifier architecture that overcomes the limitation of conventional ADP algorithms relying on samples of the input-output trajectories. A safety optimal control scheme was proposed in Qin et al. (2026c) for interconnected nonlinear

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systems. This scheme adopts a novel dynamic event-based sampling mechanism and has the ability to simultaneously handle discontinuous state constraints. Nonetheless, a common drawback in these policy iteration-based algorithms is the requirement of an initially admissible control, which is often difficult to obtain in practice for uncertain nonlinear systems.

Recently, several linearization approximation-based ADP schemes have been applied for nonlinear systems, which avoid the HJB equation in optimal controller designs. By Taylor series expansion, the nonlinear two-time-scale industrial model was linearized in Li et al. (2017), treating second-order residual error as a disturbance input. Mukherjee et al. (2021) exploited singular perturbation theory to derive some reduced-order suboptimal controllers, but this linearized mechanism is only made available for small perturbations to nonlinear dynamics. Using modified Koopman linearization, Zeng et al. (2024) incorporated nonlinear state and input terms for data-driven optimal controller design. However, the Koopman operator definition remains strongly dependent on the underlying model structure. An interpretable fuzzy system-based ADP algorithm was introduced in Juang and You (2024) for mobile robot control via a model-based actor-critic architecture. By Jacobian linearization principle, a locally optimal control strategy was presented in Manoochehri rahbar et al. (2025) for partially unknown bilinear systems. As a result, these existing publications are limited to first-order approximation, small-signal perturbation, or *a priori* knowledge of model dynamics.

Carleman embedding technique is a lifting procedure that transforms finite-dimensional nonlinear models into infinite-dimensional bilinear or linear representations without loss of fidelity (Kowalski & Steeb, 1991). Also, this way grants researchers the flexibility to truncate the model order to any desired finite value (Amini et al., 2019). Unlike these first-order linearization strategies that are only valid near its operating point, Carleman lifting technique could be leveraged to study control problem of nonlinear systems in a more wide sense. Based on Carleman linearization, an on-policy ADP algorithm was derived in Kar et al. (2023) for the optimal control problem of nonlinear systems with uncertain dynamics, but this learning algorithm exhibits three key shortcomings. Firstly, the algorithm requires knowledge of system input dynamics rather than being totally model-free. Secondly, this scheme supposes the availability of an initially admissible control, which is often impractical for uncertain nonlinear systems. Thirdly, the state trajectories of truncated system (learning-oriented) deviate from those of the original system due to neglected truncation errors. This situation induces an incorrect solution or even algorithm divergence during iteration.

Motivated by the above mentioned observations, this paper studies the optimal control problem for continuous-time nonlinear polynomial systems with uncertain dynamics. By Carleman operator, we first provide Proposition 1 to lift the nonlinear model into an infinite-dimensional bilinear form. Based on this result, it is shown that the optimal feedback can be derived from a state-dependent Riccati (SDR) equation instead of HJB equation. To execute this design in the finite space, we also introduce a truncated near-optimal control protocol and prove its closed-loop stability in Theorem 1. Then, by combining homotopy-based iteration (Chen et al., 2022) and successive approximation procedure (Aganovic & Gajic, 1995), we further present a data-driven three-phase policy iteration Algorithm 1, in which the solution is guaranteed to be unique by Lemmas 3–5, to learn the near-optimal controller in the absence of the knowledge of system dynamics and its convergence proof is given in Theorem 2. It is noted that Algorithm 1 has the ability to automatically update the initial parameters and specifies the same basis functions of neural networks to approximate all unknown functions. This differs from existing first-order linearization approaches (Li et al., 2017; Manoochehri rahbar et al., 2025) that are only valid near the operating point, the proposed controller stabilizes the closed-loop system under more large-signal perturbation. Different from the existing homotopy-based policy iteration in Chen et al. (2022), which mainly focuses on finding an initial stabilizing policy for unknown linear systems,

the homotopy mechanism proposed in this paper is designed to address the initialization difficulty of adaptive optimal control for the nonlinear system, i.e., a high-dimensional bilinear system after Carleman approximation. Unlike some existing ADP algorithms (Chen et al., 2024; Zhang et al., 2026), neural networks are no longer needed in the initialization process of Algorithm 1 and thus the computational efficiency is significantly improved. In contrast to Kar et al. (2023), Algorithm 1 takes truncation errors into account and requires neither the information of input dynamic nor an initial stabilizing control guess. This means that our proposed methodology has better convergence property and is easier to apply for real-world engineering plants.

2. Nonlinear optimal control problem

Consider the nonlinear system

$$\dot{z} = f(z) + g(z)u \quad (1)$$

where $z = [z_1, z_2, \dots, z_{n_z}]^T \in \mathbb{R}^{n_z}$ is the system state, $u = [u_1, u_2, \dots, u_{n_u}]^T \in \mathbb{R}^{n_u}$ is the control input, both z and u are supposed to be measured, $f: \mathbb{R}^{n_z} \rightarrow \mathbb{R}^{n_z}$ and $g: \mathbb{R}^{n_z} \rightarrow \mathbb{R}^{n_z \times n_u}$ are polynomial mappings of z with $f(0) = 0$. It is pursued that the precise knowledge of both $f(z)$ and $g(z)$ is not available. The to-be-minimized cost function with respect to (1) is specified as

$$J(z, u) = \int_0^\infty (z^T Q z + u^T R u) dt \quad (2)$$

where $Q = Q^T \in \mathbb{R}^{n_z \times n_z} \geq 0$ and $R = R^T \in \mathbb{R}^{n_u \times n_u} > 0$ are weight matrices. A feedback control protocol $u: \mathbb{R}^{n_z} \rightarrow \mathbb{R}^{n_u}$ is said to be admissible in a compact set $\mathcal{Z} \subset \mathbb{R}^{n_z}$ if the closed-loop system $\dot{z} = f(z) + g(z)u$ is locally asymptotically stable at the origin and the cost $J(z, u) < +\infty$ for $\forall z \in \mathcal{Z}$. Also, the function $V(z) = J(z, u)$ is referred to as value function of the control u for all $z \in \mathcal{Z}$.

Let us review the conventional strategy that solves optimal control problem and analyse its limitations. By Liberzon (2011, Section 5.1), the optimal control input that minimizes the cost function (2) is determined by

$$u^\circ(z) = -\frac{1}{2} R^{-1} g^T(z) \nabla V(z) \quad (3)$$

where $\nabla V(z)$ refers to the gradient of $V(z)$ with respect to z and satisfies the following HJB equation:

$$\nabla V^T(z) f(z) + z^T Q z - \frac{1}{4} \nabla V^T(z) g(z) R^{-1} g^T(z) \nabla V(z) = 0 \quad (4)$$

which sometimes does not have a smooth closed-form solution and its analytical solution is usually difficult to acquire. Therefore, several numerical optimization algorithms have been reported in the literature. If $V(z)$ exists, the following policy iteration technique given in Saridis and Lee (1979) is widely used to approximate it.

- 1) Find an initial admissible control input $u^{(0)}(z)$ on $\mathcal{Z}$.
- 2) Supposing $V^{(j)}(0) = 0$ and any integer $j \geq 0$, one solves $V^{(j)}(z)$ from

$$0 = (\nabla V^{(j)}(z))^T (f(z) + g(z)u^{(j)}(z)) + z^T Q z + (u^{(j)}(z))^T R u^{(j)}(z). \quad (5)$$

- 3) Update the control input by

$$u^{(j+1)}(z) = -\frac{1}{2} R^{-1} g^T(z) \nabla V^{(j+1)}(z). \quad (6)$$

In order to execute the above algorithm, an initial condition must be set to trigger policy iteration even in its data-driven versions (Jiang & Jiang, 2017; Luo et al., 2014; Possieri & Sassano, 2023; Vrabie & Lewis, 2009). But, in the absence of the information of $f(z)$ and $g(z)$, how to find such an admissible control input is a nontrivial task.

In this work, our interest is to design an adaptive optimal control u that minimizes (2) and stabilizes (1) at the origin. Inspired by Carleman embedding technique (Kowalski & Steeb, 1991), our methodology

for the rest of this paper is to lift nonlinear model (1) into an infinite-dimensional bilinear system with no loss, and provide the researchers with the freedom of truncating the order to an arbitrarily near-optimal solution obtained from an SDR equation instead of HJB equation. Then, we present a data-driven three-phase policy iteration algorithm that automatically updates all initial parameters to learn the optimal controller without requiring the knowledge of $f(z)$ and $g(z)$.

3. Carleman linearization based solution to nonlinear optimal control problem

Let us consider nonlinear polynomial system (1), where the mappings $f(z)$ and $g(z)u$ are expressed using Kronecker powers of z as follows Kowalski and Steeb (1991, Chapter 3)

$$f(z) = \sum_{k=1}^{N_f} F_k z^{[k]} \quad (7a)$$

$$g(z)u = \sum_{i=1}^{n_u} \left(G_{0,i} + \sum_{k=1}^{N_g} G_{k,i} z^{[k]} \right) u_i \quad (7b)$$

in which $F_k \in \mathbb{R}^{n_z \times (n_z)^k}$, $G_{0,i} \in \mathbb{R}^{n_z}$, $G_{k,i} \in \mathbb{R}^{n_z \times (n_z)^k}$ are constant matrices, $z^{[k]} \in \mathbb{R}^{(n_z)^k} = \underbrace{z \otimes z \otimes \dots \otimes z}_{k \text{ times}}$, and $N_f, N_g \in \mathbb{Z}^+$. Note that $\otimes$ denotes Kronecker product and $\mathbb{Z}^+$ stands for positive integers. Carleman linearization is an infinite-dimensional nonlinear map in which the elements can be represented using monomials of z . For this, we define the infinite-dimensional state

$$\bar{x}(t) = \text{col}(\bar{x}_1(t), \bar{x}_2(t), \bar{x}_3(t), \dots), \quad \bar{x}_i(t) = z^{[i]}(t) \quad (8)$$

where $\bar{x}_i \in \mathbb{R}^{(n_z)^i}$, $i \in \mathbb{Z}^+$, and $\text{col}(\cdot)$ represents a column vector by stacking the embraced vectors. In what follows, we introduce a preliminary result that is useful for formulating the corresponding infinite-dimensional system.

Proposition 1. Consider system (1) and let (7) hold, then

$$\dot{\bar{x}}_i = \sum_{j=0}^{N_f-1} A_{i+j}^i \bar{x}_{i+j} + \sum_{v=1}^{n_u} B_{i-1,v}^i z^{[i-1]} u_v + \sum_{v=1}^{n_u} \sum_{j=0}^{N_g-1} B_{i+j,v}^i \bar{x}_{i+j} u_v \quad (9)$$

for all $t \geq 0$ with

$$A_{i+j-1}^i = \sum_{l=1}^i \underbrace{(I_{n_z \times n_z} \otimes \dots \otimes F_j \otimes \dots \otimes I_{n_z \times n_z})}_{l^{\text{th}} \text{ position}},$$

$$B_{i+j-1,v}^i = \sum_{l=1}^i \underbrace{(I_{n_z \times n_z} \otimes \dots \otimes G_{j,v} \otimes \dots \otimes I_{n_z \times n_z})}_{l^{\text{th}} \text{ position}}.$$

Proof. The proof is inspired by Forets and Pouly (2017, Proposition 3.2). Applying Leibniz rule, it follows from (1), (7), and (8) that

$$\begin{aligned} \dot{\bar{x}}_i &= \dot{z} \otimes z \otimes \dots \otimes z + z \otimes \dot{z} \otimes \dots \otimes z + \dots + z \otimes z \otimes \dots \otimes \dot{z} \\ &= \sum_{l=1}^i \left(z \otimes \dots \otimes \sum_{j=1}^{N_f} F_j z^{[j]} \otimes \dots \otimes z \right) \\ &\quad + \sum_{l=1}^i \left(z \otimes \dots \otimes \sum_{v=1}^{n_u} \left(G_{0,v} + \sum_{j=1}^{N_g} G_{j,v} z^{[j]} \right) u_v \otimes \dots \otimes z \right) \end{aligned}$$

From linearity of Kronecker product, the sums of the above equation can be exchanged as

$$\dot{\bar{x}}_i = \sum_{j=1}^{N_f} \sum_{l=1}^i (z \otimes \dots \otimes F_j z^{[j]} \otimes \dots \otimes z)$$

$$\begin{aligned} &+ \sum_{v=1}^{n_u} \sum_{l=1}^i (z \otimes \dots \otimes G_{0,v} u_v \otimes \dots \otimes z) \\ &+ \sum_{v=1}^{n_u} \sum_{j=1}^{N_g} \sum_{l=1}^i (z \otimes \dots \otimes G_{j,v} z^{[j]} u_v \otimes \dots \otimes z) \\ &= \sum_{j=1}^{N_f} \sum_{l=1}^i (I_{n_z} \otimes \dots \otimes F_j \otimes \dots \otimes I_{n_z}) (z \otimes \dots \otimes z^{[j]} \otimes \dots \otimes z) \\ &+ \sum_{v=1}^{n_u} \sum_{l=1}^i (I_{n_z} \otimes \dots \otimes G_{0,v} \otimes \dots \otimes I_{n_z}) (z \otimes \dots \otimes u_v \otimes \dots \otimes z) \\ &+ \sum_{v=1}^{n_u} \sum_{j=1}^{N_g} \sum_{l=1}^i ((I_{n_z} \otimes \dots \otimes G_{j,v} \otimes \dots \otimes I_{n_z}) \\ &\quad \times (z \otimes \dots \otimes z^{[j]} u_v \otimes \dots \otimes z)) \\ &= \sum_{j=1}^{N_f} A_{i+j-1}^i z^{[i+j-1]} + \sum_{v=1}^{n_u} B_{i-1,v}^i z^{[i-1]} u_v \\ &\quad + \sum_{v=1}^{n_u} \sum_{j=1}^{N_g} B_{i+j-1,v}^i z^{[i+j-1]} u_v \\ &= \sum_{j=0}^{N_f-1} A_{i+j}^i \bar{x}_{i+j} + \sum_{v=1}^{n_u} B_{i-1,v}^i z^{[i-1]} u_v + \sum_{v=1}^{n_u} \sum_{j=0}^{N_g-1} B_{i+j,v}^i \bar{x}_{i+j} u_v \end{aligned}$$

which implies (9) to be valid. $\square$

By Proposition 1 and (8), the nonlinear polynomial system (1) can be written as an infinite-dimensional bilinear system of the form

$$\dot{\bar{x}} = \bar{A} \bar{x} + \sum_{i=1}^{n_u} (\bar{B}_{0,i} + \bar{B}_{1,i} \bar{x}) u_i \quad (10)$$

where initial condition $\bar{x}(0) = \bar{x}(z(0))$, $\bar{B}_{0,i} = \text{col}(G_{0,i}, 0, 0, \dots)$, and

$$\bar{A} = \begin{bmatrix} A_1^1 & A_2^1 & A_3^1 & \dots & A_{N_f}^1 & 0 & 0 & \dots \\ 0 & A_2^2 & A_3^2 & \dots & A_{N_f}^2 & A_{N_f+1}^2 & 0 & \dots \\ 0 & 0 & A_3^3 & \dots & A_{N_f}^3 & A_{N_f+1}^3 & A_{N_f+2}^3 & \dots \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & \end{bmatrix}$$

$$\bar{B}_{1,i} = \begin{bmatrix} B_{1,i}^1 & B_{2,i}^1 & B_{3,i}^1 & \dots & B_{N_g,i}^1 & 0 & 0 & \dots \\ B_{1,i}^2 & B_{2,i}^2 & B_{3,i}^2 & \dots & B_{N_g,i}^2 & B_{N_g+1,i}^2 & 0 & \dots \\ 0 & B_{2,i}^3 & B_{3,i}^3 & \dots & B_{N_g,i}^3 & B_{N_g+1,i}^3 & B_{N_g+2,i}^3 & \dots \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & \end{bmatrix}.$$

However, system (10) contains some redundant state variables. For example, when $n_z = 2$, by (8), we have $\bar{x}_2 = z^{[2]} = [z_1^2, z_1 z_2, z_2 z_1, z_2^2]^T$. To circumvent this issue, let

$$z^{(i)} = [z_1^i, z_1^{i-1} z_2, \dots, z_1^{i-1} z_{n_z}, z_1^{i-2} z_2^2, z_1^{i-2} z_2 z_3, \dots, z_1^{i-2} z_2 z_{n_z}, \dots, z_{n_z}^i]^T \quad (11)$$

for $i \in \mathbb{Z}^+$, where the dimension of $z^{(i)}$ is $n_i = \binom{n_z + i - 1}{i}$ and that there exist constant matrices such that

$$\begin{cases} z^{[i]} = M_i z^{(i)} \\ z^{(i)} = W_i z^{[i]} \end{cases} \quad (12)$$

with $M_i \in \mathbb{R}^{(n_z)^i \times n_i}$ and $W_i \in \mathbb{R}^{n_i \times (n_z)^i}$. We define the new infinite-dimensional state

$$x(t) = \text{col}(x_1(t), x_2(t), x_3(t), \dots), \quad x_i(t) = z^{(i)}(t) \quad (13)$$

where $x_i \in \mathbb{R}^{(n_i)}$, $i \in \mathbb{Z}^+$. Consequently, by (8), (12), and (13), we can rearrange (10) as

$$\dot{x} = \mathcal{W} \bar{A} \mathcal{M} x + \sum_{i=1}^{n_u} (\mathcal{W} \bar{B}_{0,i} + \mathcal{W} \bar{B}_{1,i} \mathcal{M} x) u_i \equiv A x + B_x u \quad (14)$$

with $A = \mathcal{W}\bar{A}\mathcal{M}$, $B_x = [B_{0,1} + B_{1,1}x, B_{0,2} + B_{1,2}x, \dots, B_{0,n_u} + B_{1,n_u}x]$, $\mathcal{W} = \text{blockdiag}\{W_1, W_2, W_3, \dots\}$, $\mathcal{M} = \text{blockdiag}\{M_1, M_2, M_3, \dots\}$, $B_{0,i} = \mathcal{W}\bar{B}_{0,i}$, $B_{1,i} = \mathcal{W}\bar{B}_{1,i}\mathcal{M}$, and $\text{blockdiag}(\cdot)$ denotes the operation of block diagonal matrix. Then, based on the resulting bilinear model (14), we can solve the nonlinear optimal control problem with respect to (1) and (2) in the Carleman state space. For this, we rewrite the cost function (2) as

$$J(x, u) = \int_0^\infty (x^T Q_g x + u^T R u) dt \quad (15)$$

where $Q_g = \text{blockdiag}\{Q, 0, 0, \dots\}$.

Applying dynamic programming to (14) and (15) results in the steady-state HJB equation

$$\nabla V^T(x) A x + x^T Q_g x - \frac{1}{4} \nabla V^T(x) B_x R^{-1} B_x^T \nabla V(x) = 0 \quad (16)$$

where the value function $V(x) = J(x, u)$. By Cebuhar and Costanza (1984), the solution to (16) can be sought in the form $\nabla V(x) = 2P(x)x$, where $P(x)$ is the symmetric matrix-valued function. As a result, substituting $\nabla V(x) = 2P(x)x$ into (16) yields the SDR equation

$$A^T P(x) + P(x) A + Q_g - P(x) B_x R^{-1} B_x^T P(x) = 0. \quad (17)$$

From the Hamiltonian, we have

$$\mathcal{H}(x, u, P(x)) = (Ax + B_x u)^T P(x)x + x^T P(x)(Ax + B_x u) + x^T Q_g x + u^T R u.$$

Hence, the necessary optimality condition $\partial \mathcal{H}(x, u, P(x)) / \partial u$ gives the optimal control protocol

$$u^*(x) = -R^{-1} B_x^T P(x)x \quad (18)$$

where $P(x)$ is determined by (17).

By Carleman embedding technique, we could employ (18) to replace the traditional nonlinear optimal control solution (3). Nonetheless, it would not be possible to construct an infinite-dimensional state to apply the optimal controller (18) in practice. As a result, one must resort to using a finite truncation of the lifted system (14), and executing the control design in the truncated space. A finite $N \in \mathbb{Z}^+$ order truncation of x is defined as

$$\chi = \text{col}(x_1, x_2, x_3, \dots, x_N) \in \mathbb{R}^{\sum_{j=1}^N n_j} \quad (19)$$

where x_j , $j = 1, 2, \dots, N$, is defined in (13). By (17) and (18), the resulting truncated near optimal control input is given by

$$u(\chi) = -R^{-1} B_\chi^T \mathcal{P}(\chi) \chi \quad (20)$$

where $\mathcal{P}(\chi)$ satisfies the following SDR equation:

$$A^T \mathcal{P}(\chi) + \mathcal{P}(\chi) A + Q - \mathcal{P}(\chi) B_\chi R^{-1} B_\chi^T \mathcal{P}(\chi) = 0 \quad (21)$$

with

$$\mathcal{W}_N = \text{blockdiag}\{W_1, W_2, \dots, W_N\}$$

$$\mathcal{M}_N = \text{blockdiag}\{M_1, M_2, \dots, M_N\}$$

$$A = \mathcal{W}_N \bar{A} \mathcal{M}_N, B_{0,i} = \mathcal{W}_N \bar{B}_{0,i}, B_{1,i} = \mathcal{W}_N \bar{B}_{1,i} \mathcal{M}_N$$

$$Q = \text{blockdiag}\{Q, 0_{\sum_{j=2}^N n_j \times \sum_{j=2}^N n_j}\}$$

$$\bar{B}_{0,i} = \text{col}(G_{0,i}, 0_{(n_z)^2}, 0_{(n_z)^3}, \dots, 0_{(n_z)^N})$$

$$\bar{B}_{1,i} = \begin{bmatrix} B_{1,i}^1 & B_{1,i}^2 & B_{1,i}^3 & B_{1,i}^4 & \dots & B_{1,i}^N \\ B_{1,i}^2 & B_{1,i}^3 & B_{1,i}^4 & B_{1,i}^5 & \dots & B_{1,i}^{N+1} \\ 0 & B_{1,i}^3 & B_{1,i}^4 & B_{1,i}^5 & \dots & B_{1,i}^{N+2} \\ 0 & 0 & B_{1,i}^4 & B_{1,i}^5 & \dots & B_{1,i}^{N+3} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & B_{1,i}^N \end{bmatrix}$$

$$B_\chi = [B_{0,1} + B_{1,1}\chi, B_{0,2} + B_{1,2}\chi, \dots, B_{0,n_u} + B_{1,n_u}\chi]$$

$$\bar{A} = \begin{bmatrix} A_1^1 & A_2^1 & A_3^1 & \dots & A_N^1 \\ 0 & A_2^2 & A_3^2 & \dots & A_N^2 \\ 0 & 0 & A_3^3 & \dots & A_N^3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & A_N^N \end{bmatrix}.$$

Assumption 1. The pair (A, B_0) is stabilizable, χ stays in the stabilizability domain $X = \{\chi \in \mathbb{R}^{\sum_{j=1}^N n_j} | (A, B_\chi) \text{ stabilizable}\}$, and the pair $(A, \sqrt{Q})$ is detectable, where $B_0 = [B_{0,1}, B_{0,2}, \dots, B_{0,n_u}]$.

Remark 1. Assumption 1 is a elementary prerequisite in the optimal control of bilinear systems (Aganovic & Gajic, 1995), which ensures the existence of the solution to the associated SDR Eq. (21). When the model parameters of system (1) are available, the stabilizability and detectability conditions can be explicitly verified using the Popov-Belevitch-Hautus test. However, when the system dynamics are completely unknown, determining whether Assumption 1 holds solely from measured data which remains a long-standing and challenging open problem in the field of adaptive optimal control.

According to the theory of Riccati equation (Bittanti et al., 1991), the solution $\mathcal{P}(\chi)$ of (21) is unique and determined under Assumption 1. Intuitively, the solution $\mathcal{P}(\chi)$ to (21) is also easier to find than $\nabla V(x)$ to (4). This means that we can operate (20) to approximate the solution of the nonlinear optimal problem associated with (1) and (2), and improve its optimality by increasing the truncation order N . The following theorem concerns the stability of the closed-loop system (1) under the truncated control protocol (20).

Theorem 1. Consider system (1) and let Assumption 1 hold. Then, there exists a truncated control input (20) that makes (1) locally asymptotically stable¹ for given truncation order $N \geq 1$.

Proof. By Proposition 1, (12), and (19), we formulate the truncated system as

$$\dot{\chi} = \mathcal{A}\chi + B_\chi u + \mathcal{A}^1 \zeta_1 + \sum_{i=1}^{n_u} B_{1,i}^1 \zeta_2 u_i \quad (22)$$

where the items $\mathcal{A}^1 \zeta_1$ and $\sum_{i=1}^{n_u} B_{1,i}^1 \zeta_2 u_i$ are truncation errors, and

$$\zeta_1 = \text{col}(x_{N+1}, x_{N+2}, \dots, x_{N+N_f-1})$$

$$\zeta_2 = \text{col}(x_{N+1}, x_{N+2}, \dots, x_{N+N_g-1})$$

$$\mathcal{M}_{N_f} = \text{blockdiag}\{M_{N+1}, M_{N+2}, \dots, M_{N+N_f-1}\}$$

$$\mathcal{M}_{N_g} = \text{blockdiag}\{M_{N+1}, M_{N+2}, \dots, M_{N+N_g-1}\}$$

$$\mathcal{A}^1 = \mathcal{W}_N \begin{bmatrix} A_{N+1}^1 & A_{N+2}^1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A_{N+1}^{N-1} & A_{N+2}^{N-1} & \dots & 0 \\ A_{N+1}^N & A_{N+2}^N & \dots & A_{N+N_f-1}^N \end{bmatrix} \mathcal{M}_{N_f}$$

$$B_{1,i}^1 = \mathcal{W}_N \begin{bmatrix} B_{N+1,i}^1 & B_{N+2,i}^1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ B_{N+1,i}^{N-1} & B_{N+2,i}^{N-1} & \dots & 0 \\ B_{N+1,i}^N & B_{N+2,i}^N & \dots & B_{N+N_g-1,i}^N \end{bmatrix} \mathcal{M}_{N_g}.$$

Let us ignore the truncation errors $\mathcal{A}^1 \zeta_1$ and $\sum_{i=1}^{n_u} B_{1,i}^1 \zeta_2 u_i$ in (22) for now. By Bittanti et al. (1991), under Assumption 1, the solution $\mathcal{P}(\chi)$ of (21) is unique and determined. Hence, the following Lyapunov candidate is considered

$$\mathcal{V}(t) = \chi^T(t) \mathcal{P}(\chi) \chi(t).$$

Then taking derivative on both sides of $\mathcal{V}(t)$, it follows from (20)–(22) that

$$\begin{aligned} \dot{\mathcal{V}}(t) &= \chi^T \left(A - B_\chi R^{-1} B_\chi^T \mathcal{P}(\chi) \right)^T \mathcal{P}(\chi) \chi + \chi^T \mathcal{P}(\chi) \left(A - B_\chi R^{-1} B_\chi^T \mathcal{P}(\chi) \right) \chi \\ &= -\chi^T \left(Q + \mathcal{P}(\chi) B_\chi R^{-1} B_\chi^T \mathcal{P}(\chi) \right) \chi. \end{aligned}$$

¹ A dynamical system is said to be locally and asymptotically stable if there exists a pair (σ, t_0) such that if $\|z(t_0)\| \leq \sigma$ then its equilibrium point $z = 0$ is asymptotically stable, i.e., there exist constants $c_1, c_2 > 0$ such that $\|z(t)\| \leq c_1 \|z(t_0)\| e^{-c_2(t-t_0)}$ for all $t \geq t_0$.

From the above equation, we can see that $\dot{V} \leq 0$ for all $\chi(t)$. Setting $\dot{V} = 0$ leads to $z = 0$. Also, by (13), we have

$$\lim_{z \rightarrow 0} z^{(i)} \rightarrow 0, \forall i \in \mathbb{Z}^+ \Leftrightarrow \lim_{z \rightarrow 0} x_i \rightarrow 0 \quad (23)$$

which, together with (19), concludes that $\dot{V} = 0$ results in $\chi = 0$. Therefore, by LaSalle's invariance principle, the system (22) under the input (20) is asymptotically stable when the higher-order items $\mathcal{A}^1 \zeta_1 \triangleq 0$ and $\sum_{i=1}^{n_u} B_{1,i}^1 \zeta_2 u_i \triangleq 0$. In what follows, we analyze the impact of truncation errors on the closed-loop stability. According to the matrix operation, there exist some bounded constant matrices $D_k \in \mathbb{R}^{(\sum_{i=1}^N n_i) \times n_{N+k}}$, $k = 1, 2, \dots, N_f - 1$, such that $\mathcal{A}^1 \zeta_1 = \sum_{k=1}^{N_f-1} D_k x_{N+k}$. By applying the triangle inequality and Lancaster and Farahat (1972, Theorem 8), we have

$$\begin{aligned} \left\| \sum_{k=1}^{N_f-1} D_k x_{N+k} \right\| &\leq \sum_{k=1}^{N_f-1} \|D_k x_{N+k}\| \\ &\leq \sum_{k=1}^{N_f-1} \|D_k\| \|z^{(N+k)}\| \\ &= \sum_{k=1}^{N_f-1} \|D_k\| \|\mathcal{M}_{N+k}\| \underbrace{\|z \otimes z \otimes \dots \otimes z\|}_{N+k} \\ &= \sum_{k=1}^{N_f-1} \|D_k\| \|\mathcal{M}_{N+k}\| \|z\|^{N+k} \end{aligned}$$

with $\mathcal{M}_{N+k} = \text{blockdiag}\{M_{N+1}, M_{N+2}, \dots, M_{N+k}\}$. As a result, it follows that

$$\lim_{z \rightarrow 0} \frac{\|\mathcal{A}^1 \zeta_1\|}{\|z\|} \leq \lim_{z \rightarrow 0} \frac{\sum_{k=1}^{N_f-1} \|D_k\| \|\mathcal{M}_{N+k}\| \|z\|^{N+k}}{\|z\|} = 0$$

which implies $\lim_{z \rightarrow 0} (\|\mathcal{A}^1 \zeta_1\| / \|z\|) = 0$ because of $N \geq 1$. Similar to the proof of the above equation, it is easily derived that the higher-order item $\sum_{i=1}^{n_u} B_{1,i}^1 \zeta_2 u_i$ also satisfies $\lim_{z \rightarrow 0} (\|\sum_{i=1}^{n_u} B_{1,i}^1 \zeta_2 u_i\| / \|z\|) = 0$. Since the closed-loop system (22) is asymptotically stable without truncation error items, by Verhulst (2012, Theorem 7.1), there exists a pair (σ, t_0) with $z(t_0) = z_0$ such that $\|z(t_0)\| \leq \sigma$ then the solution $z = 0$ is asymptotically stable for actual system (22). This reveals that, by (23), the truncated controller (20) makes (1) locally asymptotically stable. The proof is thus completed. $\square$

From the proof of Theorem 1, σ -neighborhood of $z = 0$ (a neighborhood of the origin with radius σ) for the closed-loop system (1) can be expanded by increasing the truncation order N . Unfortunately, this also brings more computational load. Hence N should be chosen reasonably on the basis of practical plants. The following section introduces how to obtain the optimal control (20) in the setting of this paper.

4. Adaptive optimal controller design: a three-phase policy iteration

The design of the truncated optimal controller (20) requires solving SDR Eq. (21) that does not have analytical solution in general. The purpose of this section is to derive a data-driven policy iteration algorithm which not only does not rely on the dynamic knowledge of (1) but also automatically update all initial parameters to design (20). We break down this section into three subsections to achieve such a goal. At the outset, we review two model-based policy iteration algorithms in the following lemmas.

Lemma 1. Suppose that Assumption 1 holds. Let $P^{(0)}$ be a stabilizing matrix, i.e., $\mathcal{A} - B_0 R^{-1} B_0^T P^{(0)}$ is Hurwitz. Solve the following Lyapunov equations for $P^{(j+1)}(\chi)$ with $j = 0, 1, 2, \dots$,

$$(\mathcal{A}^{(j)})^T P^{(j+1)}(\chi) + P^{(j+1)}(\chi) \mathcal{A}^{(j)} + Q^{(j)} = 0 \quad (24)$$

where $\mathcal{A}^{(j)} = \mathcal{A} - B_\chi R^{-1} B_\chi^T P^{(j)}(\chi)$ and $Q^{(j)} = Q + P^{(j)}(\chi) B_\chi R^{-1} B_\chi^T P^{(j)}(\chi)$. Then, the property $\lim_{j \rightarrow \infty} P^{(j)}(\chi) = P(\chi)$ holds.

Lemma 2. Suppose that the pair $(\mathcal{A}, B_0)$ is stabilizable. Let kernel matrix $M^{(0)} = 0$, scalar $\alpha^{(0)} = 0$, constant $\gamma > 0$, and constant $\beta > \max_k(\text{Re}(\lambda_k(\mathcal{A})))$.

Solve $M^{(j+1)}$ from

$$(\mathcal{A}^{(j)})^T M^{(j+1)} + M^{(j+1)} \mathcal{A}^{(j)} + Q^{(j)} = 0 \quad (25)$$

for each $j = 0, 1, 2, \dots$, where $\mathcal{A}^{(j)} = \mathcal{A} - B_0 R^{-1} B_0^T M^{(j)} - \beta I_n + \sum_{l=0}^j \alpha^{(l)} I_n$, $Q^{(j)} = Q + M^{(j)} B_0 R^{-1} B_0^T M^{(j)} + \gamma I_n$, and $0 < \alpha^{(j+1)} < -\max_k(\text{Re}(\lambda_k(\mathcal{A} - B_0 R^{-1} B_0^T M^{(j+1)} - \beta I_n + \sum_{l=0}^j \alpha^{(l)} I_n)))$, $n = \sum_{v=1}^N n_v$, and the operator $\text{Re}(\lambda_k(\cdot))$ means the real part of k th eigenvalues of matrix. Then, for each j

$$\max_k(\text{Re}(\lambda_k(\mathcal{A} - B_0 R^{-1} B_0^T M^{(j)}))) < \beta - \sum_{l=0}^j \alpha^{(l)}. \quad (26)$$

The results in Lemmas 1 and 2 are direct extensions of Aganovic and Gajic (1995, Algorithm 3.2) and Chen et al. (2022, Lemma 2), respectively. Given an initial stabilizing matrix, Lemma 1 introduces a successive approximation procedure to capture the solution $P(\chi)$ of (21). Also, Lemma 2 provides a homotopy-based strategy to find such an initial matrix.

4.1. Phase One: learning an initial stability factor using data

From Lemma 2, a sufficiently large constant β should be set during initialization to ensure $\beta > \max_k(\text{Re}(\lambda_k(\mathcal{A})))$. In fact $\beta > \max_k(\text{Re}(\lambda_k(\mathcal{A})))$ is equivalent to matrix $\mathcal{A} - \beta I_n$ being Hurwitz. Thereby, β is recommended here as the initial stability factor. In this subsection, our goal is to use system data to address the issue that the selection of β depends on the prior knowledge of $\mathcal{A}$. For this, the following Lyapunov equation is considered

$$(\mathcal{A} - \beta^{(j)} I_n)^T W + W (\mathcal{A} - \beta^{(j)} I_n) + Q + \kappa I_n = 0 \quad (27)$$

which, together with Zhou et al. (1996, Lemma 3.19), implies that $\mathcal{A} - \beta^{(j)} I_n$ is Hurwitz if $W > 0$, wherein $\kappa > 0$ and $\beta^{(j)}$ is defined as a constant which is strictly increasing and satisfies

$$\beta^{(0)} = 0, \beta^{(j)} < \beta^{(j+1)}, \lim_{j \rightarrow \infty} \beta^{(j)} = +\infty \quad (28)$$

with $j = 0, 1, 2, \dots$. As a result, by (27), we can judge $\beta^{(j)} > \max_k(\text{Re}(\lambda_k(\mathcal{A})))$ by observing whether W is positive definite and then capture $\beta \leftarrow \beta^{(j)}$. For the learning design, we rewrite system (22) as

$$\dot{\chi} = (\mathcal{A} - \beta^{(j)} I_n) \chi + \beta^{(j)} \chi + B_0 u + \sum_{i=1}^{n_u} B_{1,i} \chi u_i + \mathcal{A}^1 \zeta_1 + \sum_{i=1}^{n_u} B_{1,i}^1 \zeta_2 u_i. \quad (29)$$

Inspired by Jiang and Jiang (2017), it follows from (27) and (29) that

$$\chi^T(t) W \chi(t) - \chi^T(t_0) W \chi(t_0) \quad (30)$$

$$\begin{aligned} &= - \int_{t_0}^t \chi^T (Q + \kappa I_n) \chi d\tau + 2\beta^{(j)} \int_{t_0}^t \chi^T W \chi d\tau \\ &\quad + 2 \int_{t_0}^t u^T B_0^T W \chi d\tau + 2 \sum_{i=1}^{n_u} \int_{t_0}^t (\chi u_i)^T B_{1,i}^T W \chi d\tau \\ &\quad + 2 \int_{t_0}^t \zeta_1^T (\mathcal{A}^1)^T W \chi d\tau + 2 \sum_{i=1}^{n_u} \int_{t_0}^t (\zeta_2 u_i)^T (B_{1,i}^1)^T W \chi d\tau. \end{aligned}$$

Before proceeding, for matrices $\mathfrak{A} = [a_1, a_2, \dots, a_q] \in \mathbb{R}^{v \times q}$, $\mathfrak{B} = [b_{ij}] \in \mathbb{R}^{v \times v}$, and vector $\mathfrak{C} = [c_i] \in \mathbb{R}^v$, define the operators

$$\begin{aligned} \text{vec}(\mathfrak{A}) &\triangleq \text{col}(a_1, a_2, \dots, a_v) \in \mathbb{R}^{vq} \\ \text{ves}(\mathfrak{B}) &\triangleq [b_{11}, 2b_{12}, \dots, 2b_{1v}, b_{22}, 2b_{23}, \dots, 2b_{(v-1)v}, b_{vv}]^T \in \mathbb{R}^{0.5v(v+1)} \\ \text{vev}(\mathfrak{C}) &\triangleq [c_1^2, c_1 c_2, \dots, c_1 c_v, c_2^2, c_2 c_3, \dots, c_{v-1} c_v, c_v^2]^T \in \mathbb{R}^{0.5v(v+1)}. \end{aligned}$$

For any three vectors $\zeta_1, \zeta_2, \zeta_3$ and a sufficiently large number $l > 0$, we further define

$$\Xi_{\zeta_1, \zeta_2} = \left[\int_{t_0}^{t_1} \zeta_1 \otimes \zeta_2 d\tau, \int_{t_1}^{t_2} \zeta_1 \otimes \zeta_2 d\tau, \dots, \int_{t_{l-1}}^{t_l} \zeta_1 \otimes \zeta_2 d\tau \right]^T$$

$$\Xi_{\zeta_1, \zeta_2, \zeta_3} = \left[\int_{t_0}^{t_1} \zeta_1 \otimes \zeta_2 \otimes \zeta_3 d\tau, \int_{t_1}^{t_2} \zeta_1 \otimes \zeta_2 \otimes \zeta_3 d\tau, \dots, \int_{t_{l-1}}^{t_l} \zeta_1 \otimes \zeta_2 \otimes \zeta_3 d\tau \right]^T.$$

Based on these formulations, one changes (30) into the following data-driven form:

$$\mathcal{X}_1^{(j)} \mathcal{Z}_1 = \mathcal{Y}_1 \quad (31)$$

where

$$\mathcal{X}_1^{(j)} = [\Theta_1^{(j)}, -2\Xi_{\chi, u}, -2\Lambda]$$

$$\mathcal{Z}_1 = \text{col}(\text{vec}(W), \text{vec}(B_0^T W), \text{vec}(B_{1,1}^T W), \text{vec}(B_{1,2}^T W), \dots, \text{vec}(B_{1,n_u}^T W), \text{vec}((\mathcal{A}^1)^T W), \text{vec}((B_{1,1}^1)^T W), \text{vec}((B_{1,2}^1)^T W), \dots, \text{vec}((B_{1,n_u}^1)^T W))$$

$$\mathcal{Y}_1 = -\Xi_{\chi, \chi} \text{vec}(Q + \kappa I_n)$$

$$\Theta_1^{(j)} = [\text{vev}(\chi(t_1)) - \text{vev}(\chi(t_0)) - 2\beta^{(j)} \int_{t_0}^{t_1} \text{vev}(\chi) d\tau, \text{vev}(\chi(t_2)) - \text{vev}(\chi(t_1)) - 2\beta^{(j)} \int_{t_1}^{t_2} \text{vev}(\chi) d\tau, \dots, \text{vev}(\chi(t_l)) - \text{vev}(\chi(t_{l-1})) - 2\beta^{(j)} \int_{t_{l-1}}^{t_l} \text{vev}(\chi) d\tau]^T$$

$$\Lambda = [\Xi_{\chi, \chi u_1}, \Xi_{\chi, \chi u_2}, \dots, \Xi_{\chi, \chi u_{n_u}}, \Xi_{\chi, \zeta_1}, \Xi_{\chi, \zeta_2 u_1}, \Xi_{\chi, \zeta_2 u_2}, \dots, \Xi_{\chi, \zeta_2 u_{n_u}}].$$

Inspired by Jiang and Jiang (2017), the following lemma shows that the uniqueness of solution to (31) can be assured under a rank condition, which can be satisfied by adding probing noise to the control input.

Lemma 3. *If there exists an integer l^* such that for all $l > l^*$, for any sequence $0 \leq t_0 < t_1 < \dots < t_{l-1} < t_l$,*

$$\text{rank}(\Psi) = \frac{1}{2}n(n+1) + nn_u + n^2 n_u + np + nqn_u \quad (32)$$

with $\Psi = [\Xi_{\chi, \chi}, \Xi_{\chi, u}, \Lambda]$, $p = \sum_{k=1}^{N_f-1} n_k$, and $q = \sum_{k=1}^{N_g-1} n_k$, then (31) has an unique solution of $\mathcal{Z}_1$ for all j .

Proof. We give the proof by contradiction. Assume $\Gamma^o = \text{col}(\Omega_1^o, \Omega_2^o, \Omega_{31}^o, \Omega_{32}^o, \dots, \Omega_{51}^o, \Omega_{52}^o, \dots, \Omega_{5n_u}^o)$ is a nonzero solution to the following equation

$$0 = \mathcal{X}_1^{(j)} \Gamma^o \quad (33)$$

then there exist $\Omega_1 = \Omega_1^T \in \mathbb{R}^{n \times n}$, $\Omega_2 \in \mathbb{R}^{n_u \times n}$, $\Omega_{31} \in \mathbb{R}^{n \times n}$, $\Omega_{32} \in \mathbb{R}^{n \times n}$, $\dots$, $\Omega_{3n_u} \in \mathbb{R}^{n \times n}$, $\Omega_4 \in \mathbb{R}^{p \times n}$, $\Omega_{51} \in \mathbb{R}^{q \times n}$, $\Omega_{52} \in \mathbb{R}^{q \times n}$, $\dots$, $\Omega_{5n_u} \in \mathbb{R}^{q \times n}$, such that $\Omega_1^o = \text{vec}(\Omega_1)$, $\Omega_2^o = \text{vec}(\Omega_2)$, $\Omega_{31}^o = \text{vec}(\Omega_{31})$, $\Omega_{32}^o = \text{vec}(\Omega_{32})$, $\dots$, $\Omega_{3n_u}^o = \text{vec}(\Omega_{3n_u})$, $\Omega_4^o = \text{vec}(\Omega_4)$, $\Omega_{51}^o = \text{vec}(\Omega_{51})$, $\Omega_{52}^o = \text{vec}(\Omega_{52})$, $\dots$, $\Omega_{5n_u}^o = \text{vec}(\Omega_{5n_u})$. From (27), (29), and (30), one has

$$0 = [\Xi_{\chi, \chi}, 2\Xi_{\chi, u}, 2\Lambda] \Upsilon_1 \quad (34)$$

where $\Upsilon_1 = \text{col}(\text{vec}(\Gamma_1), T)$, $T = \text{col}(\text{vec}(\Gamma_2), \text{vec}(\Gamma_{31}), \text{vec}(\Gamma_{32}), \dots, \text{vec}(\Gamma_{3n_u}), \text{vec}(\Gamma_4), \text{vec}(\Gamma_{51}), \text{vec}(\Gamma_{52}), \dots, \text{vec}(\Gamma_{5n_u}))$, $\Gamma_1 = (\mathcal{A} - \beta^{(j)} I_n)^T \Omega_1 + \Omega_1 (\mathcal{A} - \beta^{(j)} I_n)$, $\Gamma_2 = B_0^T \Omega_1 - \Omega_2$, $\Gamma_{31} = B_{1,1}^T \Omega_1 - \Omega_{31}$, $\Gamma_{32} = B_{1,2}^T \Omega_1 - \Omega_{32}$, $\dots$, $\Gamma_{3n_u} = B_{1,n_u}^T \Omega_1 - \Omega_{3n_u}$, $\Gamma_4 = (\mathcal{A}^1)^T \Omega_1 - \Omega_4$, $\Gamma_{51} = (B_{1,1}^1)^T \Omega_1 - \Omega_{51}$, $\Gamma_{52} = (B_{1,2}^1)^T \Omega_1 - \Omega_{52}$, $\dots$, $\Gamma_{5n_u} = (B_{1,n_u}^1)^T \Omega_1 - \Omega_{5n_u}$. Let $\Theta_\chi = \left[\int_{t_0}^{t_1} \text{vev}(\chi) d\tau, \int_{t_1}^{t_2} \text{vev}(\chi) d\tau, \dots, \int_{t_{l-1}}^{t_l} \text{vev}(\chi) d\tau \right]^T$. By (34), we have

$$[\Theta_\chi, 2\Xi_{\chi, u}, 2\Lambda] \Upsilon_2 = 0$$

with $\Upsilon_2 = \text{col}(\text{vec}(\Gamma_1), T)$. Under the full rank condition of (32), we get $\Omega_2 = B_0^T \Omega_1$, $\Omega_{31} = B_{1,1}^T \Omega_1$, $\Omega_{32} = B_{1,2}^T \Omega_1, \dots, \Omega_{3n_u} = B_{1,n_u}^T \Omega_1$, $\Omega_4 = (\mathcal{A}^1)^T \Omega_1$, $\Omega_{51} = (B_{1,1}^1)^T \Omega_1$, $\Omega_{52} = (B_{1,2}^1)^T \Omega_1, \dots, \Omega_{5n_u} = (B_{1,n_u}^1)^T \Omega_1$ from above equation. Since $\mathcal{A} - \beta^{(j)} I_n$ is Hurwitz, $\Omega_1 = 0$ is the unique solution to $(\mathcal{A} - \beta^{(j)} I_n)^T \Omega_1 + \Omega_1 (\mathcal{A} - \beta^{(j)} I_n) = 0$. As a result, we have $\Gamma^o = 0$. It contradicts with the assumption of $\Gamma^o \neq 0$ in (33). This concludes the proof. $\square$

Consequently, we significantly remove the model information required in initializing β by checking W to be positive definite in (31). The following subsection is to learn an initial stabilizing matrix M and circumvent the requirement of the dynamic matrices, i.e., $\mathcal{A}$ and B_0 in (25).

4.2. Phase Two: learning an initial kernel matrix using data

In order to generalize the model-based algorithm in Lemma 2 into a data-driven version, we reformulate system (22) as

$$\dot{\chi} = \mathcal{A}^{(j)} \chi + \beta_2^{(j)} \chi + B_0(u + R^{-1} B_0^T M^{(j)} \chi) + \sum_{i=1}^{n_u} B_{1,i} \chi u_i + \mathcal{A}^1 \zeta_1 + \sum_{i=1}^{n_u} B_{1,i}^1 \zeta_2 u_i \quad (35)$$

where $\beta_2^{(j)} = \beta - \sum_{l=0}^j \alpha^{(l)}$. By (25) and (35), one has

$$\begin{aligned} & \chi^T(t) M^{(j+1)} \chi(t) - \chi^T(t_0) M^{(j+1)} \chi(t_0) \\ &= 2 \int_{t_0}^t \zeta_1^T (\mathcal{A}^1)^T M^{(j+1)} \chi d\tau + 2\beta_2^{(j)} \int_{t_0}^t \chi^T M^{(j+1)} \chi d\tau \\ &+ 2 \int_{t_0}^t (u + R^{-1} B_0^T M^{(j)} \chi)^T B_0^T M^{(j+1)} \chi d\tau \\ &+ 2 \sum_{i=1}^{n_u} \int_{t_0}^t (\chi u_i)^T B_{1,i}^T M^{(j+1)} \chi d\tau - \int_{t_0}^t \chi^T Q^{(j)} \chi d\tau \\ &+ 2 \sum_{i=1}^{n_u} \int_{t_0}^t (\zeta_2 u_i)^T (B_{1,i}^1)^T M^{(j+1)} \chi d\tau \end{aligned} \quad (36)$$

which implies the linear algebraic equation

$$\mathcal{X}_2^{(j)} \mathcal{Z}_2^{(j+1)} = \mathcal{Y}_2^{(j)} \quad (37)$$

with

$$\begin{aligned} \mathcal{X}_2^{(j)} &= [\Theta_2^{(j)}, -2\Xi_{\chi, \chi} (I_n \otimes R^{-1} B_0^T M^{(j)}) - 2\Xi_{\chi, u}, -2\Lambda] \\ \mathcal{Z}_2^{(j+1)} &= \text{col}(\text{vec}(M^{(j+1)}), \text{vec}(B_0^T M^{(j+1)}), \text{vec}(B_{1,1}^T M^{(j+1)}), \text{vec}(B_{1,2}^T M^{(j+1)}), \dots, \text{vec}(B_{1,n_u}^T M^{(j+1)}), \text{vec}((\mathcal{A}^1)^T M^{(j+1)}), \text{vec}((B_{1,1}^1)^T M^{(j+1)}), \text{vec}((B_{1,2}^1)^T M^{(j+1)}), \dots, \text{vec}((B_{1,n_u}^1)^T M^{(j+1)})) \\ \mathcal{Y}_2^{(j)} &= -\Xi_{\chi, \chi} \text{vec}(Q^{(j)}) \end{aligned}$$

$$\begin{aligned} \Theta_2^{(j)} &= [\text{vev}(\chi(t_1)) - \text{vev}(\chi(t_0)) - 2\beta_2^{(j)} \int_{t_0}^{t_1} \text{vev}(\chi) d\tau, \text{vev}(\chi(t_2)) - \text{vev}(\chi(t_1)) - 2\beta_2^{(j)} \int_{t_1}^{t_2} \text{vev}(\chi) d\tau, \dots, \text{vev}(\chi(t_l)) - \text{vev}(\chi(t_{l-1})) - 2\beta_2^{(j)} \int_{t_{l-1}}^{t_l} \text{vev}(\chi) d\tau]^T. \end{aligned}$$

Lemma 4. *If there exists an integer l^* such that for all $l > l^*$, for any sequence $0 \leq t_0 < t_1 < \dots < t_{l-1} < t_l$, the rank condition (32) holds, then (37) has an unique solution of $\mathcal{Z}_2^{(j+1)}$ for $j = 0, 1, 2, \dots$.*

Proof. Due to space constraints, we omit the proof of Lemma 4, as it is similar to that of Lemma 3. $\square$

According to Lemma 2, the formulation (37) is a homotopy-based data-driven algorithm that learns a stabilizing matrix $M^{(j)}$ when $\beta -$

$\sum_{l=0}^j \alpha^{(l)} \leq 0$. Note that the iteration length $\alpha^{(j+1)}$ is required in (37), when the iteration step moves from j to $j+1$. However, if one directly follows Lemma 2, the definition of $\alpha^{(j+1)}$ depends on the prior information of $\mathcal{A}$ and B_0 . In what follows, we suggest a model-free design for $\alpha^{(j+1)}$ to remove this requirement. For this, we rewrite (25) as

$$(\mathcal{A}^{(j+1)})^T M^{(j+1)} + M^{(j+1)} \mathcal{A}^{(j+1)} = -(\mathcal{Q}_2^{(j+1)} - 2\alpha^{(j+1)} M^{(j+1)}) \quad (38)$$

where $\mathcal{Q}_2^{(j+1)} = \mathcal{Q}^{(j)} + (B_0^T M^{(j+1)})^T R^{-1} B_0^T M^{(j+1)} + (B_0^T M^{(j+1)} - B_0^T M^{(j)})^T R^{-1} (B_0^T M^{(j+1)} - B_0^T M^{(j)})$. Meanwhile, the constraint $0 < \alpha^{(j+1)} < -\max_k (\text{Re}(\lambda_k(\mathcal{A} - B_0 R^{-1} B_0^T M^{(j+1)} - \beta I_n + \sum_{l=0}^j \alpha^{(l)} I_n)))$ in Lemma 2 means that $\mathcal{A}^{(j+1)}$ is Hurwitz. Since $M^{(j+1)} > 0$ in (25), $\mathcal{A}^{(j+1)}$ being Hurwitz reveals that $(\mathcal{Q}_2^{(j+1)} - 2\alpha^{(j+1)} M^{(j+1)}) > 0$ from (38) (Zhou et al., 1996, Lemma 3.18). Also, let the iteration length satisfy

$$0 < \alpha^{(j+1)} < \frac{\sigma_{\min}(\mathcal{Q}_2^{(j+1)})}{2\sigma_{\max}(M^{(j+1)})} \quad (39)$$

which leads to $(\mathcal{Q}_2^{(j+1)} - 2\alpha^{(j+1)} M^{(j+1)}) > 0$, where $\sigma_{\min}(\cdot)$ and $\sigma_{\max}(\cdot)$ indicate the minimum and maximum of singular values of matrix, respectively. Therefore, the model-free design for $\alpha^{(j+1)}$ given in (39) (wherein $B_0^T M^{(j+1)}$ can be obtained during iteration) is a viable alternative to the mode-based condition in Lemma 2.

4.3. Phase Three: learning an optimal control using data

From Phase One and Phase Two, the initial matrix $\mathcal{P}^{(0)}$ can be obtained through the input-state trajectory data of (1). In the following subsection, we shall present the data-driven design for (24). To begin with, we rewrite system (22) as

$$\dot{\chi} = \mathcal{A}^{(j)} \chi + B_0 v^{(j)} + \sum_{i=1}^{n_u} B_{1,i} \chi v_i^{(j)} + \mathcal{A}^1 \varsigma_1 + \sum_{i=1}^{n_u} B_{1,i}^1 \varsigma_2 u_i \quad (40)$$

where $v^{(j)} = u + R^{-1} B_\chi^T \mathcal{P}^{(j)}(\chi) \chi \stackrel{\Delta}{=} [v_1^{(j)}, v_2^{(j)}, \dots, v_{n_u}^{(j)}]^T$. It follows from (24) and (40) that

$$\begin{aligned} & \chi^T(t) \mathcal{P}^{(j+1)}(\chi) \chi(t) - \chi^T(t_0) \mathcal{P}^{(j+1)}(\chi) \chi(t_0) \\ &= - \int_{t_0}^t \chi^T \mathcal{Q}^{(j)} \chi d\tau + 2 \int_{t_0}^t (v^{(j)})^T B_0^T \mathcal{P}^{(j+1)}(\chi) \chi d\tau \\ & \quad + 2 \sum_{i=1}^{n_u} \int_{t_0}^t (\chi v_i^{(j)})^T B_{1,i}^T \mathcal{P}^{(j+1)}(\chi) \chi d\tau \\ & \quad + 2 \int_{t_0}^t \varsigma_1^T (\mathcal{A}^1)^T \mathcal{P}^{(j+1)}(\chi) \chi d\tau \\ & \quad + 2 \sum_{i=1}^{n_u} \int_{t_0}^t (\varsigma_2 u_i)^T (B_{1,i}^1)^T \mathcal{P}^{(j+1)}(\chi) \chi d\tau. \end{aligned} \quad (41)$$

Next, we employ the linear combination of basis functions, i.e., neural networks, to approximate the function $\mathcal{P}^{(j)}(\chi)$. Let $\Phi(\chi) \in \mathbb{R}^H$ represent the vector composed of linearly independent basis functions. By approximation theory (Powell, 1981), we set

$$\begin{aligned} \text{ves}(\hat{\mathcal{P}}^{(j)}(\chi)) &= \hat{L}_1^{(j)} \Phi(\chi) + e_1 \\ \text{vec}(B_0^T \hat{\mathcal{P}}^{(j)}(\chi)) &= \hat{L}_2^{(j)} \Phi(\chi) + e_2 \\ \text{vec}(B_{1,i}^T \hat{\mathcal{P}}^{(j)}(\chi)) &= \hat{L}_{3,i}^{(j)} \Phi(\chi) + e_{3,i} \\ \text{vec}((\mathcal{A}^1)^T \hat{\mathcal{P}}^{(j)}(\chi)) &= \hat{L}_4^{(j)} \Phi(\chi) + e_4 \\ \text{vec}((B_{1,i}^1)^T \hat{\mathcal{P}}^{(j)}(\chi)) &= \hat{L}_{5,i}^{(j)} \Phi(\chi) + e_{5,i} \end{aligned}$$

for each $j \in \mathbb{Z}^+$, where $\hat{L}_1^{(j)} \in \mathbb{R}^{0.5n(n+1) \times H}$, $\hat{L}_2^{(j)} \in \mathbb{R}^{n n_u \times H}$, $\hat{L}_{3,i}^{(j)} \in \mathbb{R}^{n \times H}$, $\hat{L}_4^{(j)} \in \mathbb{R}^{n p \times H}$ and $\hat{L}_{5,i}^{(j)} \in \mathbb{R}^{n q \times H}$ are the weight matrices of basis functions. It is noted that $\hat{L}_1^{(j)} \Phi(\chi)$ can be organized as $\sum_{k=1}^v \hat{L}_{1,k}^{(j)} \Phi_k(\chi)$. This can be seen as a neural network with v neurons on the hidden layer and activation functions $\Phi_k(\chi)$, and $\hat{L}_{1,k}^{(j)}$ denotes the weights of the

output layer (Vrabie & Lewis, 2009). Then, replacing $\mathcal{P}^{(j+1)}(\chi)$, $\mathcal{Q}^{(j)}$, $B_0^T \mathcal{P}^{(j+1)}(\chi)$, $B_{1,i}^T \mathcal{P}^{(j+1)}(\chi)$, $(\mathcal{A}^1)^T \mathcal{P}^{(j+1)}(\chi)$, and $(B_{1,i}^1)^T \mathcal{P}^{(j+1)}(\chi)$ in (41) with their approximations, one has

$$\begin{aligned} & (\Phi^T(\chi) \otimes \text{vec}^T(\chi)) \Big|_{t_s}^{t_{s+1}} \text{vec}(\hat{L}_1^{(j+1)}) \\ &= 2 \left(\int_{t_s}^{t_{s+1}} \Phi^T(\chi) \otimes \chi^T \otimes (\hat{v}^{(j)})^T d\tau \right) \text{vec}(\hat{L}_2^{(j+1)}) \\ & \quad + 2 \sum_{i=1}^{n_u} \left(\int_{t_s}^{t_{s+1}} \Phi^T(\chi) \otimes \chi^T \otimes (\chi \hat{v}_i^{(j)})^T d\tau \right) \text{vec}(\hat{L}_{3,i}^{(j+1)}) \\ & \quad + 2 \left(\int_{t_s}^{t_{s+1}} \Phi^T(\chi) \otimes \chi^T \otimes \varsigma_1^T d\tau \right) \text{vec}(\hat{L}_4^{(j+1)}) \\ & \quad + 2 \sum_{i=1}^{n_u} \left(\int_{t_s}^{t_{s+1}} \Phi^T(\chi) \otimes \chi^T \otimes (\varsigma_2 u_i)^T d\tau \right) \text{vec}(\hat{L}_{5,i}^{(j+1)}) \\ & \quad - \left(\int_{t_s}^{t_{s+1}} \chi^T \otimes \chi^T d\tau \right) \text{vec}(\hat{\mathcal{Q}}^{(j)}) + \hat{e}_s^{(j)} \end{aligned} \quad (42)$$

wherein $\mathcal{Q}^{(j)} = \mathcal{Q} + \hat{\mathcal{P}}^{(j)}(\chi) B_\chi R^{-1} B_\chi^T \hat{\mathcal{P}}^{(j)}(\chi)$, $\hat{\mathcal{P}}^{(0)} = \mathcal{P}^{(0)}$, $\hat{v}^{(j)} = u + R^{-1} B_\chi^T \hat{\mathcal{P}}^{(j)}(\chi) \chi \stackrel{\Delta}{=} [\hat{v}_1^{(j)}, \hat{v}_2^{(j)}, \dots, \hat{v}_{n_u}^{(j)}]^T$, and $\hat{e}_s^{(j)}$ is the approximation error. Combining (42) from $s = 0$ to $s = l$ yields the equation

$$\mathcal{X}_3^{(j)} \hat{\mathcal{Z}}_3^{(j+1)} + \hat{E}^{(j)} = \mathcal{Y}_3^{(j)} \quad (43)$$

where

$$\begin{aligned} \mathcal{X}_3^{(j)} &= [\Theta_3, -2\Lambda_3^{(j)}] \\ \hat{\mathcal{Z}}_3^{(j+1)} &= \text{col}(\text{vec}(\hat{L}_1^{(j+1)}), \text{vec}(\hat{L}_2^{(j+1)}), \text{vec}(\hat{L}_{3,1}^{(j+1)}), \text{vec}(\hat{L}_{3,2}^{(j+1)}), \\ & \quad \dots, \text{vec}(\hat{L}_{3,n_u}^{(j+1)}), \text{vec}(\hat{L}_4^{(j+1)}), \text{vec}(\hat{L}_{5,1}^{(j+1)}), \text{vec}(\hat{L}_{5,2}^{(j+1)}), \\ & \quad \dots, \text{vec}(\hat{L}_{5,n_u}^{(j+1)})) \\ \hat{E}^{(j)} &= [\hat{e}_0^{(j)}, \hat{e}_1^{(j)}, \dots, \hat{e}_l^{(j)}]^T \\ \mathcal{Y}_3^{(j)} &= -\Xi_{\chi, \chi} \text{vec}(\hat{\mathcal{Q}}^{(j)}) \\ \Theta_3 &= [(\Phi(\chi) \otimes \text{vec}(\chi))|_{t_0}^{t_1}, (\Phi(\chi) \otimes \text{vec}(\chi))|_{t_1}^{t_2}, \dots, \\ & \quad (\Phi(\chi) \otimes \text{vec}(\chi))|_{t_{l-1}}^{t_l}]^T \\ \Lambda_3^{(j)} &= [\Xi_{\Phi(\chi), \chi, \hat{v}^{(j)}}, \Xi_{\Phi(\chi), \chi, \chi \hat{v}_1^{(j)}}, \Xi_{\Phi(\chi), \chi, \chi \hat{v}_2^{(j)}}, \dots, \Xi_{\Phi(\chi), \chi, \chi \hat{v}_{n_u}^{(j)}}, \\ & \quad \Xi_{\Phi(\chi), \chi, \varsigma_1}, \Xi_{\Phi(\chi), \chi, \varsigma_2 u_1}, \Xi_{\Phi(\chi), \chi, \varsigma_2 u_2}, \dots, \Xi_{\Phi(\chi), \chi, \varsigma_2 u_{n_u}}] \end{aligned}$$

Note that the solution to (43) can be obtained in the sense of least squares, i.e., $\hat{\mathcal{Z}}_3^{(j+1)} = ((\mathcal{X}_3^{(j)})^T \mathcal{X}_3^{(j)})^{-1} (\mathcal{X}_3^{(j)})^T \mathcal{Y}_3^{(j)}$.

Assumption 2. Given $H \in \mathbb{Z}^+$, there exist $l^* \in \mathbb{Z}^+$ and $\mu > 0$ such that for all $l > l^*$ and $j = 0, 1, 2, \dots$,

$$\frac{1}{l} (\mathcal{X}_3^{(j)})^T \mathcal{X}_3^{(j)} \geq \mu I. \quad (44)$$

Lemma 5. Consider (43) and let Assumption 2 hold. Given an initial stabilizing matrix $\mathcal{P}^{(0)}$, for any $\varepsilon > 0$ and each $j \in \mathbb{Z}^+$, there exists an $H^* \in \mathbb{Z}^+$, such that if $H > H^*$, following inequalities hold

$$\begin{aligned} & \|\hat{\mathcal{P}}^{(j)}(\chi) - \mathcal{P}^{(j)}(\chi)\| \leq \varepsilon \\ & \|B_0^T \hat{\mathcal{P}}^{(j)}(\chi) - B_0^T \mathcal{P}^{(j)}(\chi)\| \leq \varepsilon \\ & \|B_{1,i}^T \hat{\mathcal{P}}^{(j)}(\chi) - B_{1,i}^T \mathcal{P}^{(j)}(\chi)\| \leq \varepsilon, \quad i = 1, 2, \dots, n_u. \end{aligned}$$

Proof. Let $E^{(j)} = [e_0^{(j)}, e_1^{(j)}, \dots, e_l^{(j)}]^T$ denote the true truncation errors. It follows from (43) that $\mathcal{X}_3^{(j)} \hat{\mathcal{Z}}_3^{(j+1)} + E^{(j)} = \mathcal{Y}_3^{(j)}$, where $\hat{\mathcal{Z}}_3^{(j+1)}$ is the approximate solution of $\mathcal{Z}_3^{(j+1)}$. Then, we obtain

$$\hat{E}^{(j)} = \mathcal{X}_3^{(j)} \hat{\mathcal{Z}}_3^{(j+1)} + E^{(j)}$$

where $\hat{\mathcal{Z}}_3^{(j+1)} = \mathcal{Z}_3^{(j+1)} - \hat{\mathcal{Z}}_3^{(j+1)}$ is the approximation error. Since the solution is found by the least-squares method, one has $\frac{1}{l} (\hat{E}^{(j)})^T \hat{E}^{(j)} \leq$

$\frac{1}{l}(E^{(j)})^T E^{(j)}$. Besides, it is noted that

$$\begin{aligned} & \frac{1}{l} \left(\tilde{z}_3^{(j+1)} \right)^T \left(\chi_3^{(j)} \right)^T \chi_3^{(j)} \tilde{z}_3^{(j+1)} \\ &= \frac{1}{l} \left(\hat{E}^{(j)} - E^{(j)} \right)^T \left(\hat{E}^{(j)} - E^{(j)} \right) \leq \frac{4}{l} \left(E^{(j)} \right)^T E^{(j)}. \end{aligned}$$

Hence, combining the above equation and [Assumption 2](#) yields

$$\begin{aligned} \left(\tilde{z}_3^{(j+1)} \right)^T \tilde{z}_3^{(j+1)} &\leq \frac{4}{\mu l} \left(E^{(j)} \right)^T E^{(j)} \\ &\leq \frac{4}{\mu} \max_{0 \leq s \leq l} \left(e_s^{(j)} \right)^2. \end{aligned}$$

For any $0 \leq s \leq l$, $e_s^{(j)}$ converges to zero as $H \rightarrow \infty$. As a result, by the above equation, we have $\lim_{H \rightarrow \infty} \left(\tilde{z}_3^{(j+1)} \right)^T \tilde{z}_3^{(j+1)} = 0$, which implies that the estimation of weight matrices $\hat{L}_1^{(j)}$, $\hat{L}_2^{(j)}$, and $\hat{L}_{3,i}^{(j)}$, $i = 1, 2, \dots, n_u$, converge to their true values as $H \rightarrow \infty$. The proof is completed. $\square$

Algorithm 1 Data-driven three-phase policy iteration.

- 1: **Result:** The learned optimal controller $u = -R^{-1}B_\chi^T \mathcal{P}(\chi)\chi$
 - 2: Apply any input u to explore system (1) and collect the data such that (32) and (44) hold
 - 3: **Phase One:** Initial stability factor design
 - 4: Set $W = 0$ and $j = 0$
 - 5: **while** $W \leq 0$ **do**
 - 6: Choose $\beta^{(j)}$ via (28)
 - 7: Solve $\mathcal{Z}_1$ from (31)
 - 8: $j \leftarrow j + 1$
 - 9: **end while**
 - 10: **Phase Two:** Initial kernel matrix design
 - 11: Set $\beta = \beta^{(j)}$, $M^{(0)} = 0$, $\alpha^{(0)} = 0$ and reset $j = 0$
 - 12: **while** $\sum_{i=0}^j \alpha^{(i)} < \beta$ **do**
 - 13: Solve $\mathcal{Z}_2^{(j+1)}$ from (37)
 - 14: Choose $\alpha^{(j+1)}$ via (39)
 - 15: $j \leftarrow j + 1$
 - 16: **end while**
 - 17: **Phase Three:** Optimal control protocol design
 - 18: Set $\hat{\mathcal{P}}^{(0)} = M^{(j)}$, $B_\chi^T \hat{\mathcal{P}}^{(0)} = B_\chi^T M^{(j)}$, $\Delta = I$, choose a small threshold $\delta > 0$, and reset $j = 0$
 - 19: **while** $\|\Delta\| > \delta$ **do**
 - 20: Solve $\hat{\mathcal{Z}}_3^{(j+1)}$ from (43)
 - 21: $\Delta = \hat{L}_1^{(j+1)} - \hat{L}_1^{(j)}$
 - 22: $j \leftarrow j + 1$
 - 23: **end while**
 - 24: **Return:** The convergent matrices $B_0^T \mathcal{P}(\chi) = \text{vec}^{-1}(\hat{L}_2^{(j)} \Phi(\chi))$ and $B_{1,i}^T \mathcal{P}(\chi) = \text{vec}^{-1}(\hat{L}_{3,i}^{(j)} \Phi(\chi))$, $i = 1, 2, \dots, n_u$
-

The developed three-phase algorithm is shown in [Algorithm 1](#). If $\mathcal{A}$ is inherently Hurwitz, we can directly execute Phase Three by setting $\hat{\mathcal{P}}^{(0)} = 0$ to learn the optimal controller. Also, since the exploratory input u differs from the updated control $u^{(j)} = -R^{-1}B_\chi^T \mathcal{P}^{(j)}(\chi)\chi$, [Algorithm 1](#) is off-policy ([Sutton & Barto, 2018; Wang et al., 2024](#)).

Theorem 2. Consider [Algorithm 1](#) and let [Assumption 1](#) hold. For any $\varepsilon > 0$, there exist integers $j > 0$ and $H^* > 0$, such that if $H > H^*$, the following inequalities hold

$$\begin{aligned} \|\hat{\mathcal{P}}^{(j)}(\chi) - \mathcal{P}(\chi)\| &\leq \varepsilon \\ \|B_0^T \hat{\mathcal{P}}^{(j)}(\chi) - B_0^T \mathcal{P}(\chi)\| &\leq \varepsilon \\ \|B_{1,i}^T \hat{\mathcal{P}}^{(j)}(\chi) - B_{1,i}^T \mathcal{P}(\chi)\| &\leq \varepsilon, \quad i = 1, 2, \dots, n_u. \end{aligned}$$

Proof. The designed solutions in Phase One and Phase Two are unique and determined via [Lemmas 2–4](#). Under [Assumption 1](#), there exists a state-dependent solution $\mathcal{P}(\chi)$ to (21) ([Bittanti et al., 1991](#)). By [Lemma 1](#), there exists $j > 0$, such that $\|\mathcal{P}^{(j)}(\chi) - \mathcal{P}(\chi)\| \leq \varepsilon/2$. According to [Lemma 5](#), there exists $H^* > 0$, such that if $H > H^*$, $\|\hat{\mathcal{P}}^{(j)}(\chi) -$

$\mathcal{P}^{(j)}(\chi)\| \leq \varepsilon/2$. Hence, by the triangle inequality, one has

$$\|\hat{\mathcal{P}}^{(j)}(\chi) - \mathcal{P}(\chi)\| \leq \|\hat{\mathcal{P}}^{(j)}(\chi) - \mathcal{P}^{(j)}(\chi)\| + \|\mathcal{P}^{(j)}(\chi) - \mathcal{P}(\chi)\| \leq \varepsilon.$$

Similarly, it can be shown that $\|B_0^T \hat{\mathcal{P}}^{(j)}(\chi) - B_0^T \mathcal{P}(\chi)\| \leq \varepsilon$ and $\|B_{1,i}^T \hat{\mathcal{P}}^{(j)}(\chi) - B_{1,i}^T \mathcal{P}(\chi)\| \leq \varepsilon$, $i = 1, 2, \dots, n_u$. $\square$

Accordingly, we see from [Theorem 2](#) that the feedback control input designed by [Algorithm 1](#) converges to its target value in (20) as both number of basis functions of neural networks and iteration step of algorithm approaches to infinity. The nonlinear optimal control problem with uncertain dynamics is thus solved by the proposed Carleman linearization based data-driven methodology.

Remark 2. The proposed three-phase policy iteration algorithm differs fundamentally from the value iteration (VI) approach developed in [Bian and Jiang \(2022\)](#). Although both methods remove the requirement of an initially admissible control policy, they employ different learning mechanisms. The VI algorithm directly updates the value function by solving the HJB equation, which generally exhibits relatively slow convergence and requires repeated approximation of the nonlinear value function. In contrast, the proposed method first transforms the nonlinear optimal control problem into a SDR equation through Carleman approximation and then employs a homotopy-based policy iteration scheme to automatically construct an initial stabilizing controller. As a result, the proposed algorithm preserves the fast convergence property of policy iteration while avoiding the initialization difficulty encountered in conventional methods ([Vrabie & Lewis, 2009](#)).

Remark 3. The proposed methodology is developed for nonlinear polynomial systems, where the dynamics can be represented by a finite polynomial expansion as in (7). For general nonlinear systems, however, the corresponding Taylor expansion consists of an infinite number of polynomial terms. Although a finite-order truncation can still be employed to construct a finite-dimensional lifted model, the neglected higher-order terms introduce an infinite-dimensional truncation error. Such an error is further propagated through the policy iteration process and inevitably leads to additional learning approximation errors in the obtained controller. Therefore, the convergence and stability analysis established in this paper can no longer be directly applied to general nonlinear systems, which will be investigated in our future research.

5. Case studies

In this section, we shall discuss two examples to corroborate the efficiency of the developed methodology. In order to appraise the performance of the design control protocol, we further compare these results with HJB [Eq. \(4\)](#) that brings us an optimal solution.

5.1. Scalar nonlinear system

Consider the following nonlinear system

$$\dot{z} = -0.1z + zu \quad (45)$$

where $z \in \mathbb{R}$, $u \in \mathbb{R}$ denote the system state and control input, respectively. The cost function is set as

$$J = \int_0^\infty (z^2 + u^2) dt.$$

For comparison purpose, the optimal control protocol obtained from solving HJB (4) is given by $u^o = 0.1 - \sqrt{0.01 + z^2}$. The simulated results are shown in [Figs. 1–3](#), where the closed-loop system state trajectories, stability regions, the control protocols designed by the introduced method with different truncation lengths ($N = 2, 4, 6$) are provided. It can be seen that increasing the truncation order enhances the approximation of the developed control protocol to the optimal input signal u^o and enlarges the stability region of the closed-loop system. When increasing the truncation order N , the closed-loop response constructed by applying our scheme is also closer to the result of optimal input u^o .

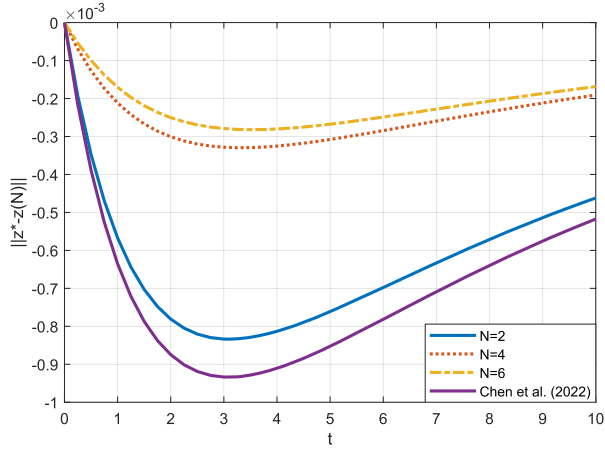


Fig. 1. Closed-loop system response with different truncation lengths and Chen et al. (2022).

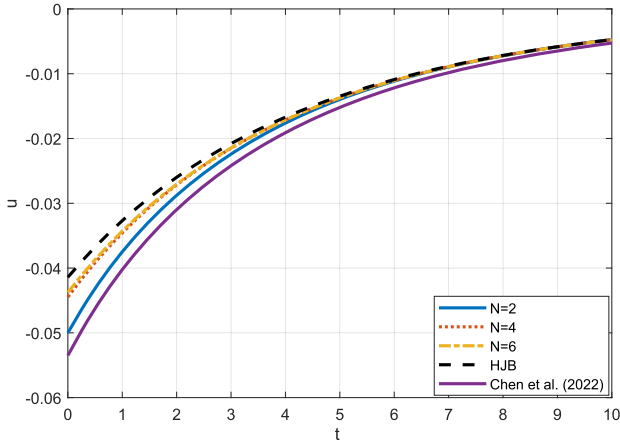


Fig. 2. Comparison of control protocols with different truncation lengths and Chen et al. (2022).

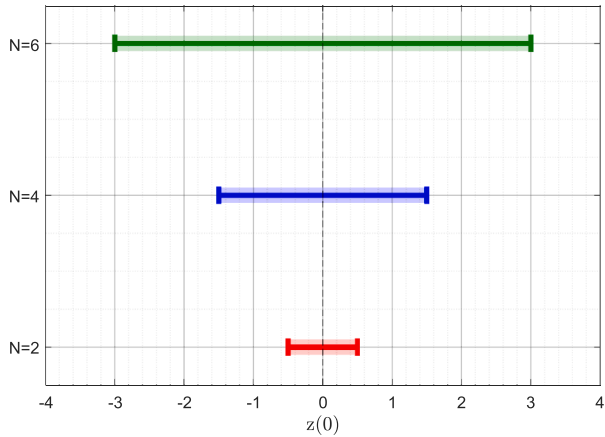


Fig. 3. Stability regions under different truncation lengths.

In addition, we provide the simulation results of Chen et al. (2022) in Figs. 1 and 2. In fact, since the homotopy-based method in Chen et al. (2022) is developed for linear systems, the control policy learned by its algorithm is essentially equivalent to that obtained by the proposed method with the truncation order set to $N = 1$. This observation is also consistent with the simulation results.

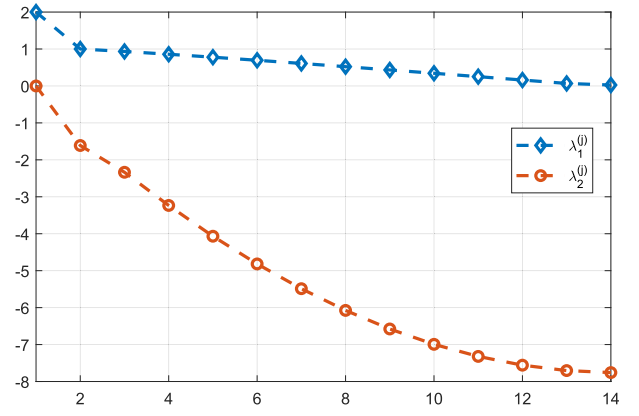


Fig. 4. Iteration process in Phase Two: Evolution of $\lambda_1^{(j)}$ and $\lambda_2^{(j)}$ with $\lambda_1^{(j)} = \beta - \sum_{i=0}^j \alpha^{(i)}$ and $\lambda_2^{(j)} = \max(\text{Re}(\lambda_k(A - B_0 R^{-1} B_0^T M^{(j)})))$.

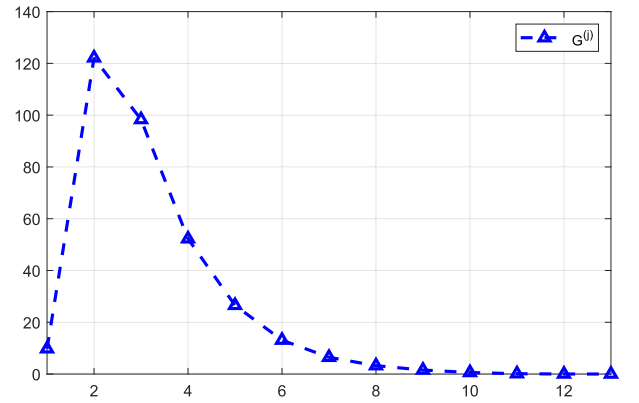


Fig. 5. Iteration process in Phase Three: Convergence of $\hat{l}_1^{(j)}$ with $G^{(j)} = \|\hat{l}_1^{(j)} - \hat{l}_1^{(j-1)}\|$.

5.2. Van der Pol oscillator

In this example, we investigate the Van der Pol oscillator that has a popular history of use in both biological and physical subjects. Following Amini et al. (2019), a specific model of the oscillator system is given by

$$\begin{cases} \dot{z}_1 = z_2 \\ \dot{z}_2 = \frac{1}{2} z_1^2 z_2 + z_1 z_2 - z_1 + (z_1 + 1)u \end{cases} \quad (46)$$

which is an unstable system, where z_1 is the position coordinate and z_2 is the velocity coordinate. The cost function to be minimized is defined as $J(z, u) = \int_0^\infty (z_2^2 + u^2) d\tau$. For this case, the optimal control input has an analytical solution to the corresponding HJB (4), namely, $u^o = -(1 + x_1)x_2$.

We choose the truncation order $N = 2$. By Proposition 1, (14), and (22), the truncated system with respect to (46) can be reformulated as the form

$$\dot{\chi} = \mathcal{A}\chi + (\mathcal{B}_{0,1} + \mathcal{B}_{1,1}\chi)u + \mathcal{A}^1\zeta_1$$

where $\chi = [z_1, z_2, z_1^2, z_1 z_2, z_2^2]^T$, $\zeta_1 = [z_1^3, z_1^2 z_2, z_1 z_2^2, z_2^3, z_1^4, z_1^3 z_2, z_1^2 z_2^2, z_1 z_2^3, z_2^4]^T$, and

$$\mathcal{A} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$$

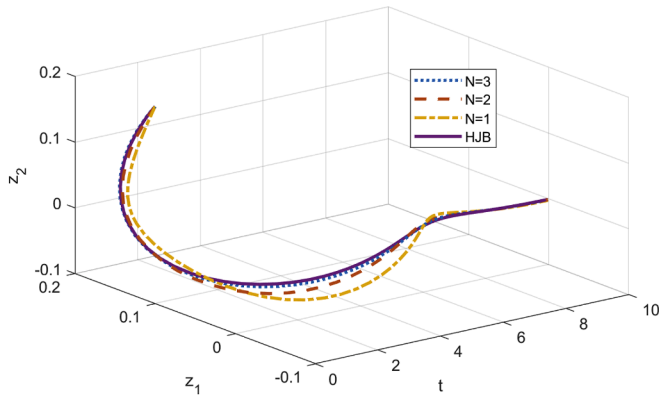


Fig. 6. Evolution of closed-loop system states: Comparison of the designed controller under $N = 1, 2, 3$ with the HJB solution.



Fig. 7. Comparison of absolute values of stable boundaries under different truncation lengths.

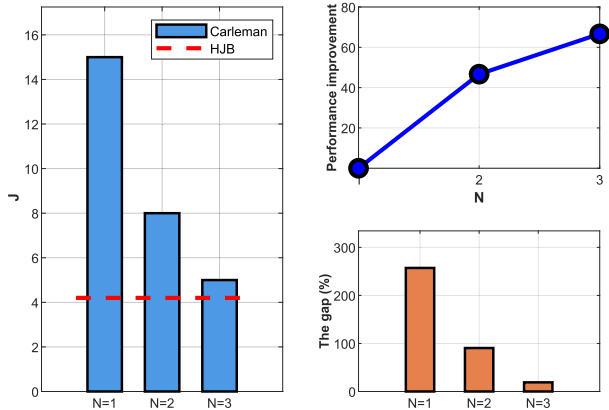


Fig. 8. Comparison of cost functions under different truncation lengths.

$$B_{0,1} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, B_{1,1} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \end{bmatrix}$$

$$A^1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

In order to execute [Algorithm 1](#), we apply a bounded exploration noise $30 \sum_{k=1}^{40} \sin(w_k t)$ with w_k being randomly sampled from $[-9, 9]$ as input signal to collect data. For the truncation order $N = 2$, the constant $\beta = 2$ ($\beta > \max(\text{Re}(\lambda_k(A)))$) is firstly obtained by phase-one learning after 1 iteration. Then a stabilizing matrix M , i.e., $A - B_0 R^{-1} B_0^T M$ is Hurwitz, is acquired by phase-two learning after 14 iterations. It can be seen from [Fig. 4](#) that the maximum real part of eigenvalues of $A - B_0 R^{-1} B_0^T M^{(j)}$ reduce from zero to negative as the term $\beta - \sum_{i=0}^j \alpha^{(i)}$ tends to zero. As depicted in [Fig. 5](#), the iteration process of Phase Three is convergent after 13 iterations. [Fig. 6](#) shows the comparison of the introduced method with different truncation lengths ($N = 1, 2, 3$) to the HJB solution for the evolution of closed-loop system states. [Fig. 7](#) illustrates that the stability region expands as the truncation order increases. It is shown in [Fig. 8](#) that, as the truncation order N increases, the control cost of the Carleman approximate solution gradually approaches that of the HJB optimal solution, and it is already very close to the global optimum when $N = 3$. We see from these simulation results that increasing the truncation order enhances the approximation performance of the developed controller to the optimal input signal u^o .

6. Conclusions

This paper has studied the adaptive optimal control problem for continuous-time nonlinear systems. *The first major contribution* of this paper is that we have transformed the nonlinear optimal control problem into a bilinear optimal control problem using Carleman embedding technique. The resulting optimal feedback controller is related to SDR equation, which effectively circumvents the traditional difficulty in obtaining the solution to HJB equation. *The second major contribution* of this paper is to present a novel three-phase policy iteration algorithm based on finite samples of input and state data. This algorithm is rendered completely model-free to find the adaptive optimal controller through three phases, which significantly strengthens its applicability for real-world engineering plants. We believe that the proposed scheme can serve as a basic framework to address various nonlinear optimal control problems, for example, tracking control, output regulation, and zero-sum game.

CRedit authorship contribution statement

Jianguo Zhao: Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Zhijiang Gao:** Writing – review & editing, Formal analysis; **Chunyu Yang:** Supervision, Methodology, Data curation; **Pak Kin Wong:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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